

Series and infinite products – 2

How fast the terms must decay: Abel, Dirichlet, Cesàro

Homework

12 сентября 2026 года

Theoretical minimum. Summation by parts turns $\sum a_n b_n$ into a boundary term plus $\sum (a_{n+1} - a_n) B_n$, and produces the tests of Dirichlet (monotone $a_n \rightarrow 0$, bounded B_n) and of Abel (monotone bounded a_n , convergent $\sum b_n$). A convergent series with monotone terms satisfies $na_n \rightarrow 0$; without monotonicity nothing of the kind holds. If $\sum a_n/n$ converges, the Cesàro means of (a_n) tend to zero, and bounded Cesàro means give convergence of $\sum a_n/n^{1+\varepsilon}$ for every positive ε . By Abel–Dini there is neither a slowest divergent nor a fastest convergent series. Finally, the Cauchy criterion in the form “all long blocks are small” is what one uses when the hypothesis speaks about sums of consecutive terms.

1. (IMC) Show that $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} \sin(\ln n)}{n^\alpha}$ converges if and only if $\alpha > 0$.
2. (SEEMOUS) Let (a_n) be a monotone decreasing sequence of real numbers converging to 0. Prove that $\sum \frac{a_n}{n}$ converges if and only if the sequence $(a_n \ln n)$ is bounded and the series $\sum (a_n - a_{n+1}) \ln n$ converges.
3. (IMC 2018) Let $(a_n)_{n \geq 0}$ be a sequence of real numbers such that $a_0 = 0$ and $a_{n+1}^3 = a_n^2 - 8$ for $n = 0, 1, 2, \dots$. Prove that the series $\sum |a_{n+1} - a_n|$ is convergent.
4. (МФТИ 1997) Prove that $\lim_{x \rightarrow +\infty} \sum_{n=1}^{\infty} \frac{nx}{(n^2+x)^2} = \frac{1}{2}$.
5. Prove Kronecker’s lemma in its general form: if $\sum \frac{a_n}{b_n}$ converges and $0 < b_n \uparrow \infty$, then $\frac{a_1 + \dots + a_n}{b_n} \rightarrow 0$. Show by an example that the monotonicity of (b_n) cannot be dropped.

6. Let $a_n \geq 0$ and let $\sum a_n$ converge. Prove that $\sum a_n^{\frac{n}{n+1}}$ converges as well, and decide whether $\sum a_n^{1-\varepsilon}$ must converge for a fixed $\varepsilon > 0$.
7. Suppose that (a_n) is bounded and that $\sum \frac{a_n}{n}$ converges. Does it follow that the Cesàro means of (a_n) converge? Does it follow that $\sum \frac{a_n}{n^\beta}$ converges for every $\beta > 1$, and what happens for $\beta < 1$?
8. Prove that a series $\sum a_n$ converges if and only if for every increasing sequence of indices the corresponding block sums tend to zero, and use this to settle the following: if the sums of every three consecutive terms tend to zero and the sums of every five consecutive terms tend to zero, must $a_n \rightarrow 0$?
9. Let $a_n \downarrow 0$. Prove that the signs $\varepsilon_n = \pm 1$ can always be chosen so that $\sum \varepsilon_n a_n$ converges, and prove the converse statement: if $\sum \varepsilon_n a_n$ converges for *every* choice of signs, then $\sum |a_n| < \infty$.
10. (RMO 1998) Let $(a_n)_{n \geq 1}$ be a sequence of real numbers such that the sequence $x_n = \sum_{k=1}^n a_k^2$ converges while the sequence $y_n = \sum_{k=1}^n a_k$ is unbounded. Prove that the sequence $b_n = \{y_n\}$ of fractional parts is divergent.